



Pakistan.AI

ISIC-2024 SLICE-3D Skin Cancer Detection

Final Project · Neural Networks

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Melanoma Triage From Total-Body Photography

- Melanoma is highly survivable when excised early; mortality rises sharply once it metastasises.
- The clinical bottleneck is **triage**: ranking which lesions warrant dermatologist review.
- ISIC-2024 frames this as a ranking task over lesions imaged by **3D total-body photography** (TBP), not dermoscopy.
- Each lesion is a small square **crop** plus per-lesion tabular metadata.

Objective: rank a patient's lesions so the **highest-risk are reviewed first**, using only signals a routine total-body scan already captures.

DATASET

SLICE-3D: 401,059 Lesion Crops

401,059

lesion crops

393

malignant

0.098%

prevalence

1,042

patients

Representative lesion crops by class

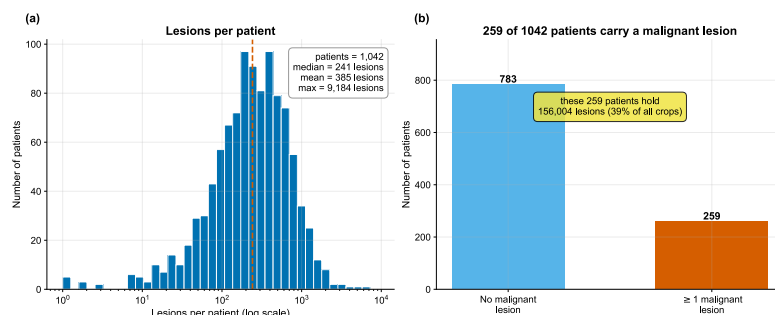


Malignant lesions trend larger, more colour-heterogeneous and more border-irregular; benign nevi are smaller and more uniform. Each crop is annotated with its clinical longest diameter.

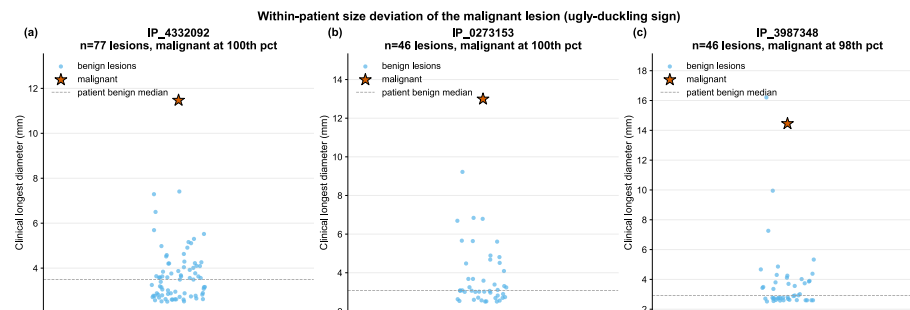
- Square TBP tiles, 61–239 px (median 131 px).
- 393 malignant against **400,666** benign.
- One cancer per **~1,021** crops.
- Head/neck site carries ~7× the baseline malignant rate.

What the Data Reveals

Patient structure — correlated lesions per patient

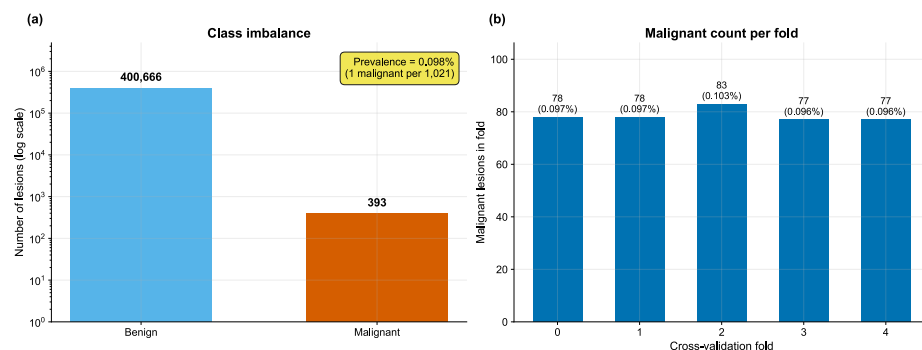


Ugly-duckling — the malignant lesion tops its patient's own size distribution



- Lesions are **patient-grouped**: 1,042 patients, median 241 crops each; the 259 malignant-carrying patients hold 89% of all crops.
- Patient-relative deviations** (ugly-duckling) carry ~65% of GBDT gain; 45.7% of cancers fall in the top 10% of their own patient's lesion sizes.
- Head/neck site runs at **~7×** the baseline malignant rate despite being the smallest site — a strong site prior.
- Hue **tbp_iv_H** alone reaches univariate AUC 0.81; malignant lesions are ~2× larger than benign.

Class Imbalance and the Metric



Class counts (log scale) and the 77–83 malignant cases per fold.

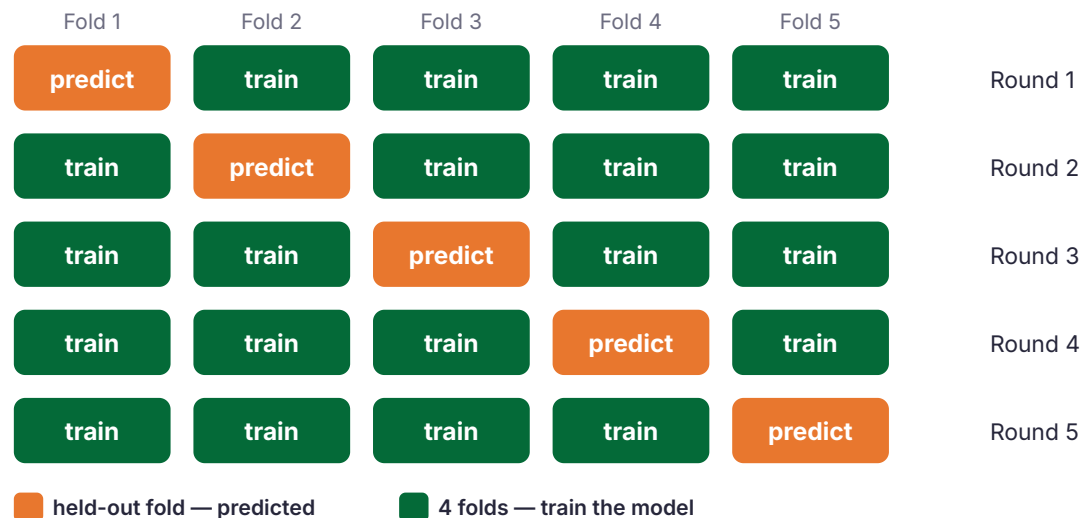
- At 0.098% prevalence, plain AUC and accuracy reward the wrong region of the curve.
- Scoring is restricted to the **high-sensitivity tail**: TPR from 0.80 to 1.0.
- Range **[0, 0.20]**: random ≈ 0.02 , perfect = 0.20.

$$\text{pAUC}_{80} = \int_{0.80}^{1.0} (1 - \text{FPR}) d(\text{TPR})$$

Official ISIC-2024 metric. Every number here is OOF pAUC@80%TPR on the same frozen folds.

Patient-Grouped, Target-Stratified 5-Fold

5-fold rotation → out-of-fold (OOF) predictions



Concatenate the 5 held-out fifths = OOF:
 one prediction per lesion. from a model that never saw it.

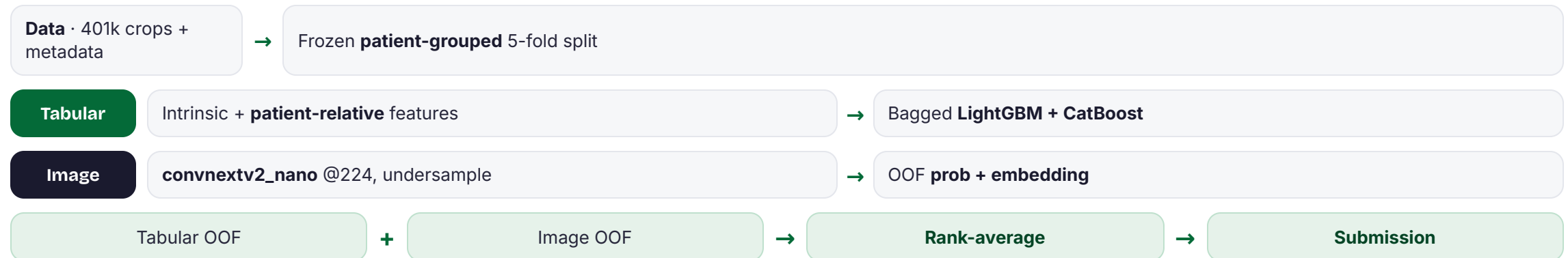
- **StratifiedGroupKFold:** grouped by patient, stratified on the label.
- **No patient straddles two folds.**
- One frozen **folds.parquet**; every model reads the identical split.
- Each fold holds 77–83 positives, so none is starved of signal.

WHY IT MATTERS

If a patient's moles appear in both train and validation, the model memorises the person, not the disease — the source of the competition's public-to-private shake-ups.

APPROACH

Two Experts and a Rank-Average Stack



Tabular expert carries most of the score; the image expert contributes a complementary signal; the combiner is a zero-parameter rank average. No external or synthetic data.

- Intrinsic features: lesion geometry, colour, size ratios.
- **Patient-relative** deviations encode the ugly-duckling sign — a lesion that departs from its patient's own moles.
- 392 of 393 cancers share a patient with benign moles, so within-patient deviation is computable for nearly every positive.
- Bagged **LightGBM** ensembled with **CatBoost**; early-stop on pAUC@80%TPR.

Table 1. Tabular LightGBM progression (OOF, leak-verified).

Configuration	OOF pAUC@80%TPR
Step 0 — broken: is_unbalance + AUC early-stop	0.09941
Step 1 — fixed early-stopping / objective	0.11826
Step 2 — wide patient-relative features	0.14420
Step 3 — bagged + cross feats + CatBoost	0.16890

Patient-relative features supply 6 of the top 7 by gain and account for ~65% of total feature importance.

- Small pretrained backbones; ImageNet weights carry no skin-cancer labels, preserving the no-external-data claim.
- Per-epoch **negative undersampling** against 0.098% prevalence.
- BCE with label smoothing, classical **transV2 augmentation**, light **EMA**, and mixup.
- Each backbone emits an OOF malignancy probability plus an embedding, stacked into the GBDT.

Resolution and light regularisation on a small backbone is the sweet spot: convnextv2_nano gains **+0.005** from 128 to 224 px; stronger EMA over-smooths and hurts.

Image backbones (OOF pAUC@80%TPR).

Backbone	pAUC
ConvNeXtV2-nano @224	0.15821
ConvNeXtV2-tiny @224	0.15824
ConvNeXtV2-nano @128	0.15311
EfficientViT-b0 @128	0.13706
ViT-tiny @128	0.13654
MobileNetV4-small @128	0.11242

nano @224 matches tiny @224 at half the cost; chosen as the image expert.

Rank-Average Stack

$$\hat{s}_i = \frac{1}{2} \left(r(p_i^{\text{tab}}) + r(p_i^{\text{img}}) \right), \quad r(p_i) = \frac{\text{rank}(p_i)}{N} \in (0, 1]$$

Worked example — ranks rescue an under-ranked cancer

Lesion	GBDT prob → rank	CNN prob → rank	avg rank	final
L1	0.91 → 5	0.40 → 3	4.0	2nd
L2 ★	0.62 → 4	0.88 → 5	4.5	1st
L3	0.30 → 3	0.55 → 4	3.5	3rd
L4	0.12 → 2	0.20 → 2	2.0	4th
L5	0.05 → 1	0.10 → 1	1.0	5th

★ L2 is the true cancer.
GBDT under-ranks it (4th); the CNN ranks it 1st.
The average rank rescues L2 to 1st place —
neither expert alone put it on top.

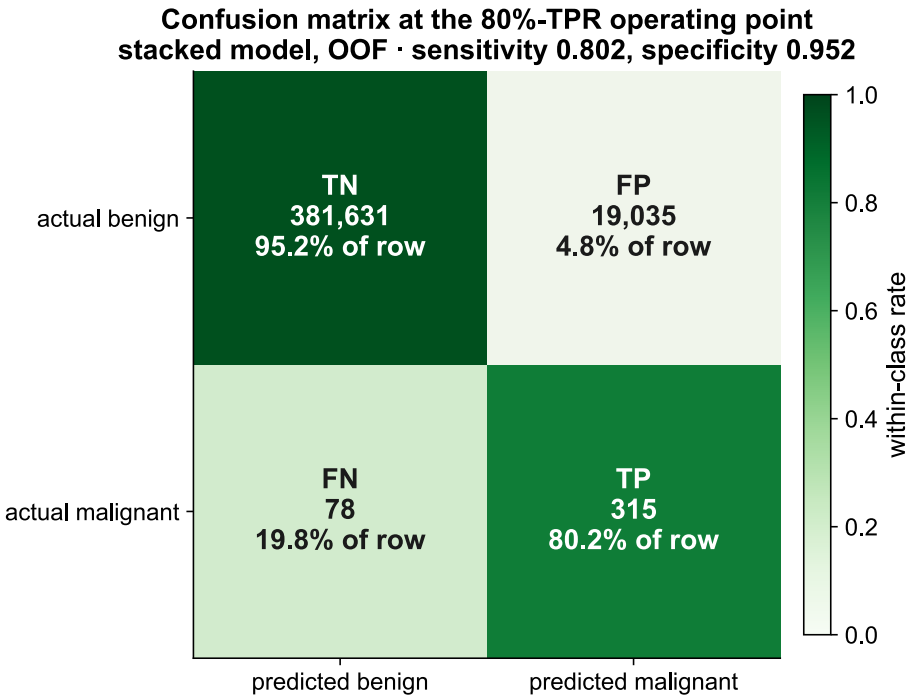
- Each expert's scores are rank-transformed, then equally averaged.
- Tabular and image make **complementary errors**; the average sits above both on every fold.
- Tabular 0.16890 + image 0.15821 → stack **0.17376**.

WHY TRIVIAL WINS

A rank average has **zero learned parameters**, so it cannot overfit the 393 positives. A learned meta-stacker reaches only 0.17108; a learned per-lesion gate falls to 0.15007 — below tabular alone.

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Ranking and the Operating Point



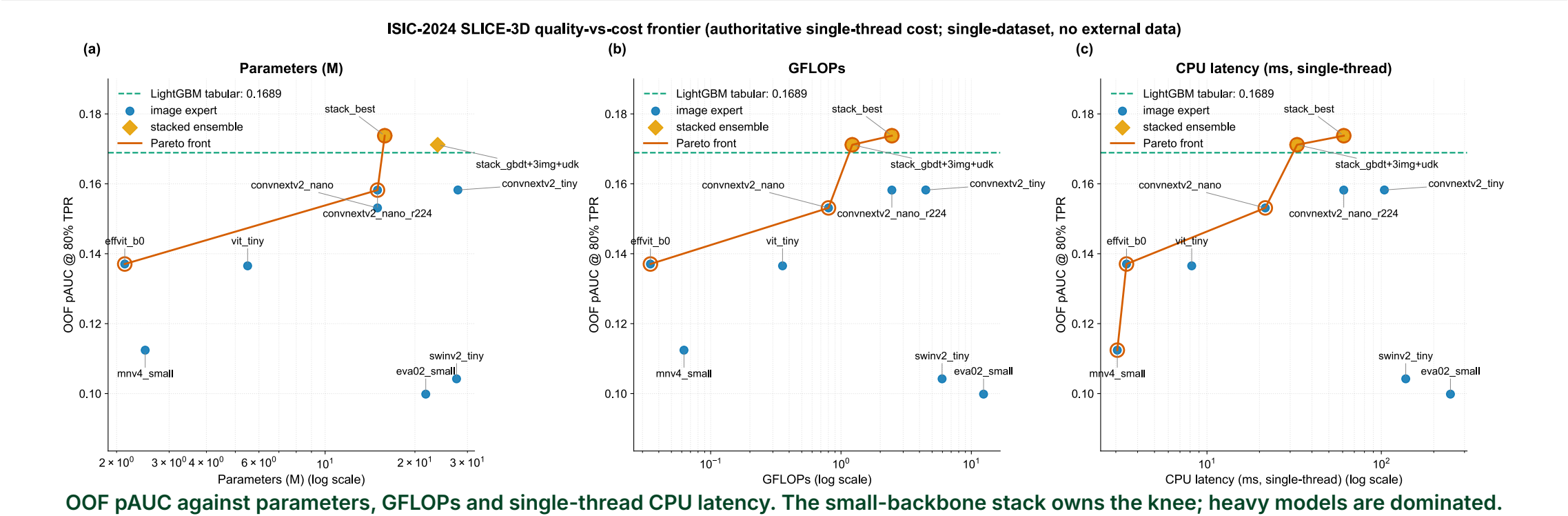
Stack, out-of-fold

pAUC @ 80% TPR (max 0.20)	0.17376
ROC-AUC	0.9666
Average precision	0.094
Sensitivity / specificity @ op. point	0.802 / 0.952

ROC-AUC is high, but at **0.098% prevalence** precision is mathematically forced low — **which is why scoring is pAUC@80%TPR**, not precision or accuracy.

Illustrative 80%-TPR point: 315/393 caught at 95.2% specificity. The score is pAUC@80%TPR — rank-based over the high-sensitivity band — so at 0.1% prevalence a large absolute false-positive count is intrinsic; a stronger model has fewer at the same sensitivity.

Quality vs. Cost: Three Axes



Model	pAUC	Params (M)	GFLOPs	CPU (ms)	Pareto
Stack (rank-avg: GBDT + nano@224)	0.17376	15.84	2.455	60.91	Yes
Tabular GBDT (LightGBM + CatBoost)	0.16890	0.86	0.000	0.02	Yes
EVA-02-small @336 (heavy ViT, dominated)	0.09984	21.74	12.409	249.18	—

Negative Results and Leaderboard Comparison

Table 3. Added complexity that did not improve OOF pAUC@80%TPR.

Idea tried	Its pAUC	Beaten by	Honest finding
Learned per-lesion gate (MoE)	0.15007	0.16890 (tabular)	loses to tabular alone
Meta-LGBM stacker	0.17108	0.17376 (rank-avg)	loses to rank-average
+ PCA image embeddings into GBDT	< 0.17376	0.17376	embeddings hurt
Heavy backbones (SwinV2 / EVA-02)	0.104 / 0.100	0.15821 (nano@224)	dominated, overfit at 393 pos
EMA 0.999 (over-smoothing)	~0.146	0.15821 (EMA0.995)	stronger EMA hurts

Every complexity increment lost on the frozen folds: each learned addition overfits the 393 positives or is dominated on cost.

Table 4. Comparison to top Kaggle solutions.

Solution	Ext.	Syn.	pAUC
1st — Novoselskiy (EVA-02 + GBDT)	Yes	Yes	0.17264
2nd — uchiyama33	Yes	Yes	—
3rd — kyohei-123	Yes	Yes	—
Ours (single-dataset)	No	No	CV 0.17376

Champions used banned external dermoscopy and ~30k synthetic positives; their own ablation reports synthetic data added only +0.0007. Ours is OOF CV, not private LB.

Constraints and Limitations

Self-imposed constraints

- **No external data.** SLICE-3D only — no ISIC-2019/2020, no PAD-UFES, no external dermoscopy.
- **No synthetic lesions.** Fabricated pathology is a label-validity hole in a medical task.
- ImageNet-pretrained encoders allowed: no skin-cancer labels, so the no-external-data property holds.

Limitations

- Only 393 positives bound model capacity; heavy backbones overfit.
- OOF pAUC is 0.17376; allowing the usual public-to-private drop, we estimate private ≈ 0.16 .
- Precision at the operating point is low by construction at this prevalence.

Conclusion

- A small, cheap stack reaches **0.17376** OOF pAUC on a leak-verified patient-grouped split.
- Patient-relative tabular features carry most of the score.
- A zero-parameter rank average beats every learned combiner.
- The result holds **without external or synthetic data**.

CLAIM

SOTA among single-dataset, no-external-data, no-synthetic solutions, reported on a quality-versus-cost frontier rather than a single leaderboard point.

At 393 positives, the efficient choices were not only cheaper — they were **more accurate**.

Attribution and Lineage

Leaderboard solutions positioned against

- **1st — I. Novoselskiy**: EVA-02 + EdgeNeXt fused with a GBDT stack; used external dermoscopy and ~30k synthetic positives (both banned here).
 - **2nd — uchiyama33**: image + tabular ensemble — source of the EMA + mixup recipe adopted for our small backbone.
 - **3rd — kyohei-123**: image/tabular blend.
- ## Public-notebook lineage (reused with attribution)
- **greysky** — LightGBM hyperparameters and the tabular-first thesis; the bagging + undersampling structure.
 - **snnclsr** — tabular feature-engineering patterns and CV structure.
 - **motono0223** — small-backbone image-baseline recipe.
 - **andreasbis** — the stacking idea: CNN OOF probability as a GBDT feature.
 - Official pAUC scorer (Kurtansky / MSKCC) vendored verbatim to verify the metric.

Dataset, Metric and Methods

Dataset & metric

- › Kurtansky, N. R. et al. *The SLICE-3D dataset: 400,000 skin-lesion crops from 3D TBP for skin-cancer detection*. Scientific Data (2024). doi:10.1038/s41597-024-03743-w
- › International Skin Imaging Collaboration. *SLICE-3D 2024 Challenge Dataset*. doi:10.34970/2024-slice-3d (CC BY-NC 4.0).
- › ISIC Research / Kurtansky (MSKCC). *Official ISIC-2024 scorer* — partial AUC above 80% TPR, scores in [0, 0.20].

Backbones & gradient boosting

- › Woo, S. et al. *ConvNeXt-V2 / FCMAE*. CVPR (2023) — the primary backbone (convnextv2_nano).
- › Ke, G. et al. *LightGBM*. NeurIPS (2017).
- › Prokhorenkova, L. et al. *CatBoost*. NeurIPS (2018).
- › Wightman, R. *timm (PyTorch Image Models)*. (2019) — backbones and the latency-axis timings.



Thank you

<https://junaidaliop.github.io/isic2024-tbp/>
Questions?